

Open AI APIs:

OpenAI provides APIs so developers can use AI in applications.

Examples:

- Chatbots in websites
- Code Assistants in VS Code

Types of OpenAI APIs:

- Chat API
- Embedding API
- Image API
- Audio API
- Moderation API

1) Chat/Completion API:

The main purpose of Chat API is used to Convert text, Answers questions, Generates Code

Example Code:

```
const openai = new OpenAI({ apiKey: 'YOUR_API_KEY' });  
const response = await openai.chat.completions.create({  
  model: 'gpt-4',  
  messages: [{ role: 'user', content: 'Explain SQL joins' }]  
});  
console.log(response.choices[0].message.content);
```

2) Embeddings API:

Used for semantic search and recommendations

Flow: Text → Vector → Compare → Result

3) Image API:

It is used to generate images

Prompt → Model → Generated Image

4) Audio API:

Used to convert Speech-to-Text/Text-to-Speech

Speech → Text or Text → Speech

Example:

audio file → API → text output

5) Moderation API:

It is used to Detect harmful or unsafe content

Input → Check content → Safe/Unsafe

Example:

- Chat filtering
- Content safety
- Social platforms

Key Parameters:

- Model
- Messages
- Temperature
- Max_tokens
- Frequency_penalty
- Presence_penalty

How to Call OpenAI APIs:

Calling an OpenAI API means sending a request from our application to OpenAI's server and getting a response.

Steps:

Here are few steps for calling OpenAI APIs:

Step 1: Get API Key

- Create an account on OpenAI
- Generate an **API key**

This is like a password to access OpenAI

Step 2: Prepare Request

Here we need to send

- Model
- Input
- Parameters

Step 3: Send Request to API

Our app sends an HTTP request to OpenAI

It includes:

- API Key
- Input Data

Step 4: Processing

OpenAI model receives our request → Understands Input → Generates Response

Step 5: Get Response

- API sends back Answer / text / output
- Our app displays it to the user

Flow: User Input → Application → OpenAI API → Model → Response → Display Output